The docking score implemented in this repository

Differentiable ZDOCK — `src/zdock`

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Abstract

This note states, in closed form, the scoring function evaluated by `zdock.score.docking_score_elec` and maximised by `zdock.search.docking_search`, and maps every symbol onto the primary literature: the pairwise shape-complementarity term of Chen & Weng (2003) [1], the interface atomic-contact-energy pair potential of Mintseris *et al.* (2007) [3], and the combination with electrostatics of Chen *et al.* (2003) [2], with the earlier grid-based formulation of Chen & Weng (2002) [4] given only where it explains a symbol that survives in the code. It also records where the implementation deviates from the papers, and why.

1 Total score

Let R denote the receptor (held fixed at the origin) and L the ligand, placed by a rigid transform $(q, \mathbf{t})$: a rotation q drawn from a finite quaternion set, followed by a translation $\mathbf{t}$ on the FFT lattice. All three terms are evaluated by scattering atoms onto a Cartesian grid of spacing h and contracting grid functions, so that the translational scan is a single FFT correlation per rotation — the construction of Katchalski-Katzir *et al.* [6], used by every ZDOCK version [4, 1, 2]. The total is

$$S(q,\mathbf{t}) \;=\; \alpha\, S_{\mathrm{PSC}}(q,\mathbf{t}) \;+\; S_{\mathrm{IFACE}}(q,\mathbf{t}) \;+\; \beta\, S_{\mathrm{ELEC}}(q,\mathbf{t}), \tag{1}$$

and **higher is better**: `docking_search` returns the top-N poses by S. Section 5 explains why keeping all three terms in a "higher is better" convention requires an explicit sign flip on S_{IFACE}.

Grid. $h = 1.2$ Å throughout, the value used by every ZDOCK paper ("A grid spacing of 1.2 Å is used throughout this study" [1]). An atom is a *surface* atom if its solvent-accessible area, computed with a 1.4 Å water probe, exceeds 1 Å^2; otherwise it is a *core* atom ([1], Methods; same definition in [4]).

2 Shape complementarity: PSC

2.1 Grid functions

Following Chen & Weng [1], Eq. (3), each molecule is described by one complex grid function. Write $\mathcal{S}$ and $\mathcal{C}$ for the sets of surface and core atoms, r_j for the van der Waals radius of atom j, and $\mathbf{x}_c$ for the centre of grid cell c. Define the occupancy sets

$$\mathrm{surf}(c) \iff \exists j \in \mathcal{S} : \|\mathbf{x}_c - \mathbf{x}_j\| < r_j, \tag{2}$$
$$\mathrm{core}(c) \iff \exists j \in \mathcal{C} : \|\mathbf{x}_c - \mathbf{x}_j\| < r_j, \tag{3}$$

with core taking precedence, and let $\mathrm{open}(c) \iff \neg\,\mathrm{surf}(c) \wedge \neg\,\mathrm{core}(c)$. Note that PSC, unlike the earlier GSC, defines *no* solvent-accessible surface layer: "any grid point that does not correspond to an atom is in the open space" [1].

The imaginary channel is the same for both partners and encodes the clash penalty,

$$\operatorname{Im}[R_{\mathrm{PSC}}(c)] = \operatorname{Im}[L_{\mathrm{PSC}}(c)] = \begin{cases} \rho & \operatorname{surf}(c) \\ \rho^2 & \operatorname{core}(c) \\ 0 & \operatorname{open}(c), \end{cases} \tag{4}$$

while the real channel differs between the two and encodes the favourable atom-pair count,

$$\operatorname{Re}[R_{\mathrm{PSC}}(c)] = \begin{cases} \big|\{\, j \in R : \|\mathbf{x}_c - \mathbf{x}_j\| < D + r_j \,\}\big| & \operatorname{open}(c) \\ 0 & \text{otherwise}, \end{cases} \tag{5}$$

$$\operatorname{Re}[L_{\mathrm{PSC}}(c)] = \big|\{\, k \in L : c = \operatorname{nearest}(\mathbf{x}_k) \,\}\big|. \tag{6}$$

2.2 Score

Eq. (4) of [1] takes the *real part* of the cross-correlation only:

$$S_{\mathrm{PSC}}(\mathbf{t}) = \operatorname{Re}\Big[\textstyle\sum_c R_{\mathrm{PSC}}(c)\, L_{\mathrm{PSC}}(c+\mathbf{t})\Big] = \underbrace{\sum_c \operatorname{Re}[R]\operatorname{Re}[L]}_{\text{favourable}} - \underbrace{\sum_c \operatorname{Im}[R]\operatorname{Im}[L]}_{\text{clash penalty}}. \tag{7}$$

The first sum is exactly the number of receptor–ligand atom pairs within the cutoff, because $\operatorname{Re}[L]$ places one count at each ligand atom and $\operatorname{Re}[R]$ reports how many receptor atoms lie within $D + r_j$ of that point. The second sum contributes

overlap	value	$\rho = 3.5$
surface–surface	$-\rho^2$	-12.25
surface–core	$-\rho^3$	-42.88
core–core	$-\rho^4$	-150.06

so deeper interpenetration is punished super-linearly, “with a higher score indicating better shape complementarity” [1].

Why this term dominates. On clash-free surface-placed decoys the bare pair count $\sum_{ij} n_{ij}$ separates near-native from non-native poses with a Mann–Whitney AUC of 0.998, i.e. it is by far the strongest single signal in the score. Omitting it — as this port did until the present revision — makes every contact a net penalty and drives the search towards poses that do not touch at all.

3 Desolvation: the IFACE pair potential

Mintseris *et al.* [3] replace ZDOCK’s averaged ACE desolvation by a pairwise statistical potential over $k = 12$ optimised atom types. With n_{ij} the number of contacts between receptor atoms of type i and ligand atoms of type j, their Eq. (11) and Eq. (14) read

$$e_{ij} = \ln \frac{n_{ij}}{c_{\text{surface-fraction};ij}}, \qquad E_{\mathrm{IFACE}} = \sum_{i=1}^{k} \sum_{j=1}^{k} e_{ij}\, n_{ij}. \tag{8}$$

On the grid (their Eq. (13)) the receptor side pre-multiplies the potential,

$$W_i(c) = \sum_j e_{ij}\, H_j(c), \qquad H_j(c) = \big|\{\, m \in R \text{ of type } j : \|\mathbf{x}_c - \mathbf{x}_m\| < 6\,\text{Å} \,\}\big|, \tag{9}$$

the 6 Å contact sphere being the one that defines n_{ij} in [3], Eqs. (8)–(9), and the ligand side is the per-type occupancy indicator of their Eq. (12), $L_i(c) \in \{0,1\}$, so that

$$S_{\mathrm{IFACE}}(\mathbf{t}) = \sigma \sum_i \sum_c W_i(c)\, L_i(c+\mathbf{t}), \qquad \sigma = -1. \tag{10}$$

4 Electrostatics

Chen & Weng [4] write the electrostatic energy as a correlation between the receptor's electric potential and the ligand's atomic charges, using CHARMM19 partial charges, with grid points in the receptor core assigned zero potential ”to avoid the contributions from non-physical receptor-core/ligand contacts”; Chen *et al.* [2], Eq. (2), carry the same term into ZDOCK through the imaginary channel. Here it is kept as a separate real correlation:

$$V(c) = \sum_{m \in R} \frac{q_m}{\|\mathbf{x}_c - \mathbf{x}_m\|} \quad (\|\cdot\| < 8\,\text{Å},\ \mathrm{open}(c)), \qquad S_{\mathrm{ELEC}}(\mathbf{t}) = \sum_c V(c)\, Q_L(c+\mathbf{t}). \tag{11}$$

5 Sign conventions

The two ingredient papers disagree about which direction is favourable, and Chen *et al.* [2] say so explicitly:

> “positive PSC scores indicate good shape complementarity … whereas ACE scores can be positive (unfavorable) or negative (favorable) … To make these two scores compatible, we flip the signs of the PSC scores.”

They resolve it by flipping PSC and *minimising*. Because the FFT search here selects poses by `topk`, this implementation flips the other term instead: $\sigma = -1$ in Eq. (10) turns the ACE-convention pair potential into a “higher is better” quantity, and every term in Eq. (1) then agrees with S_{PSC}. Omitting this flip makes S_{IFACE} anti-correlated with pose quality (measured AUC 0.127; the negated term gives 0.873), and since $|S_{\mathrm{IFACE}}| \approx |S_{\mathrm{PSC}}|$ it then dominates the total.

6 Parameters

symbol	meaning	initial value	source
h	grid spacing	1.2 Å	[1], Methods
D	PSC pair cutoff offset	3.6 Å	[1], “Optimizing PSC”
ρ	PSC clash scale	3.5	[2], Eq. (2)
α	weight on S_{PSC}	1.0	see below
β	weight on S_{ELEC}	3.0	[2], p. 82
e_{ij}	12×12 pair potential	IFACE table	[3], Eq. (11)
q_m	partial charges (11 types)	CHARMM19	[4], “Electrostatics”
σ	pair-term sign	-1	[2], p. 81

ρ, D, α, β, e_{ij} and q_m are all exposed as differentiable tensors, so any of them can be optimised; the values above are the initialisation. Note that α has no counterpart in Eq. (2) of [2], where the scale of the PSC term is fixed by ρ and the term enters with unit weight; the $\alpha = 0.01$ carried in earlier revisions of this code belongs to the *grid*-based formulation of [4], Eq. (6), whose second term is an averaged ACE energy on a kcal/mol scale rather than a pair count.

7 Where each equation comes from

here	quantity	source	source equation
(1)	$S = \alpha S_{\text{PSC}} + S_{\text{IFACE}} + \beta S_{\text{ELEC}}$	[2]	Eq. (2)
(4)	clash channel ρ, ρ^2	[1]	Eq. (3), Im
(5)	receptor pair-count channel	[1]	Eq. (3), Re[R]
(6)	ligand nearest-grid channel	[1]	Eq. (3), Re[L]
(7)	$S_{\text{PSC}} = \text{Re}[R \cdot L]$	[1]	Eq. (4)
(8)	e_{ij}, E_{IFACE}	[3]	Eqs. (11), (14)
(10)	grid form of the pair sum	[3]	Eqs. (12), (13)
$\sigma = -1$	sign reconciliation	[2]	p. 81
$D = 3.6$ Å	pair cutoff offset	[1]	"Optimizing PSC"
$\rho = 3.5$	clash scale	[2]	Eq. (2)
$\beta = 3$	electrostatics weight	[2]	p. 82
$h = 1.2$ Å	grid spacing	[1]	Methods

The four ZDOCK papers are in `ref/` of this repository (file names in the bibliography); [5, 6, 7] are cited for provenance and are not needed to run the code.

8 Deviations from the papers

1. **Rotational sampling.** ZDOCK uses evenly distributed Euler angle sets — 1800 orientations at $\Delta = 20°$, 54 000 at 6°, 180 000 at 4° ([1], "Rotational Sampling"). This implementation draws uniform random quaternions, so for the same count the worst-case angular error is larger. Finer is not automatically better: [1] report that "finer angular intervals tend to rank the best hits lower than coarser angular intervals", $\Delta = 4°$ giving the *lowest* success rate for $N_P > 100$.

2. **Ligand occupancy in S_{IFACE}.** Eq. (12) of [3] is a binary indicator (their Eq. (14) uses "≅", acknowledging that the grid form only approximates $\sum_{ij} e_{ij} n_{ij}$), which under-counts when two ligand atoms of the same type share a cell. That is negligible at $h = 1.2$ Å (cell volume 1.7 Å^3) but not at coarser spacings; the quantity is 37% smaller at $h = 3.0$ Å.

3. **Cell-volume normalisation.** The clash channel counts grid cells, so it scales as h^{-3}. `sc_cell_volume_factor` rescales it to the 1.2 Å reference, making the term spacing-invariant; it is the identity at the default spacing.

4. **DockQ.** Pose quality is scored against [7] by an atom-level approximation without residue or backbone metadata, and without symmetry handling, so the 0.23 acceptance threshold is not identical to canonical CAPRI. Ligand RMSD is reported alongside as a cross-check.

9 Measured effect of completing PSC

Twelve interface-deleaked PINDER complexes, default parameters (no training), 1900 uniform rotations, $h = 1.2$ Å, top-2000 search output, nothing injected into the candidate set. "Ceiling" is the DockQ of the best sampled rotation at the lattice-snapped native translation — a parameter-free upper bound on what the search could return; it is ≥ 0.23 for 100% of these complexes throughout.

score implementation	recall	best DockQ	s@1	s@10	s@100
clash penalty only, $h = 3.0$ Å	0.0%	0.107	0%	0%	0%
clash penalty only, $h = 1.2$ Å	0.0%	0.066	0%	0%	0%
+ pair-term sign flip	16.7%	0.116	0%	0%	0%
full PSC, Eqs. (5)–(7)	**66.7%**	**0.649**	**17%**	**42%**	**50%**

References

[1] R. Chen and Z. Weng. *A novel shape complementarity scoring function for protein–protein docking.* Proteins **51**(3), 397–408 (2003). doi:10.1002/prot.10334.
`ref/Chen_Weng_2003.pdf`

[2] R. Chen, L. Li and Z. Weng. *ZDOCK: an initial-stage protein-docking algorithm.* Proteins **52**(1), 80–87 (2003). doi:10.1002/prot.10389.
`ref/Chen_et_al_2003.pdf`

[3] J. Mintseris, B. Pierce, K. Wiehe, R. Anderson, R. Chen and Z. Weng. *Integrating statistical pair potentials into protein complex prediction.* Proteins **69**(3), 511–520 (2007). doi:10.1002/prot.21502.
`ref/Mintseris_et_al_2007.pdf`

[4] R. Chen and Z. Weng. *Docking unbound proteins using shape complementarity, desolvation, and electrostatics.* Proteins **47**(3), 281–294 (2002). doi:10.1002/prot.10092.
`ref/Chen_Weng_2002.pdf`

[5] C. Zhang, S. Vasmatzis, J. L. Cornette and C. DeLisi. *Determination of atomic desolvation energies from the structures of crystallized proteins.* J. Mol. Biol. **267**(3), 707–726 (1997). doi:10.1006/jmbi.1997.0859.

[6] E. Katchalski-Katzir, I. Shariv, M. Eisenstein, A. A. Friesem, C. Aflalo and I. A. Vakser. *Molecular surface recognition: determination of geometric fit between proteins and their ligands by correlation techniques.* Proc. Natl. Acad. Sci. USA **89**(6), 2195–2199 (1992). doi:10.1073/pnas.89.6.2195.

[7] S. Basu and B. Wallner. *DockQ: a quality measure for protein–protein docking models.* PLoS ONE **11**(8), e0161879 (2016). doi:10.1371/journal.pone.0161879.